OPTIMIZATION STUDY

Fixed-hardware optimization candidate

Revision Saved optimization from 2026-10-30; report regenerated with all failures retained · plate-frame frame · x44-standard motors

Named input case · design_baseline.json

2026-10-30T22:07:59+00:00

CALCULATED route timing: BWR/motor-limited straight profiles and retained empirical point-turn equations; not hardware calibrated. Legacy traction/stability checks remain active. Deposition and relocation retain their reference timing.

Calculated route timing: traversal_time_one_way is inactive. Speed and acceleration inputs are upper bounds. Deposition and fresh-site relocation retain reference timing.

BWR soil source-range transfer flags: 2 across the two route legs. The published soil proxy is uncalibrated for this rover; see model provenance and the BWR documentation.

Actuator offsets are measured from the drum axle: positive side is outward along the arm; positive top is above the arm. The straight tube extension is included in mass and loads. The normal bracket is assumed rigid and massless; its mass, strength and interference remain unverified.

Actuator packaging is not enforced. Its reserve and catalog endpoints are diagnostics only; they do not affect optimization or feasibility.

Constraints excluded from optimizer: dig_traction_margin, turn_traction_margin.

These checks are still calculated and included in full model feasibility. Optimizer success with a failed excluded check is not a feasible design.

1 modeled checks need attention

75 / 76 physical model checks pass · 7 capabilities not assessed

Predictions are conditional on the failed checks below. These are predictions from the stated inputs, not measured performance.

Dry rover mass	Payload per cycle	Complete cycles	Delivered volume
43.3 kg	**104.7 kg**	**3**	**188.4 L**
Before adding regolith	Total carried by four drums	Within 15 minutes; no partial delivery credit	Complete-cycle mission estimate

Construction score

Continuous mission estimate using the configured mass and energy weights. The complete-cycle estimate is reported in Performance & mass.

Construction points	Mass points	Energy points
1,692.78	**1,404.36**	**288.42**
Weighted mass + energy points	Dry-mass productivity component	Consumed-energy productivity component

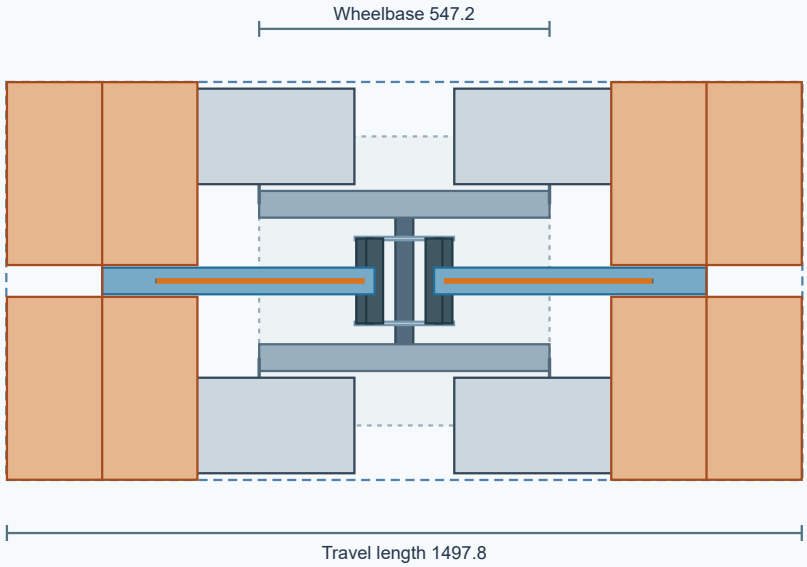
01 / ROVER CONFIGURATION

Geometry at the evaluated inputs

Level-ground schematics and one prescribed obstacle pose. Check statuses below determine whether envelopes and clearance requirements are satisfied.

Top-down view / travel pose

Orthographic footprints; front is +x (right). Dimensions in mm.



Wheel-center track: 544.7 | Overall width: 749.2 | Wheel width: 180.0

Each drum: 359.0 diameter x 344.6 axial length | Pair gap: 60.0

Blue: one arm per drum pair · orange: actuators · charcoal: aluminum cross-tubes · slate: rockers and axle.

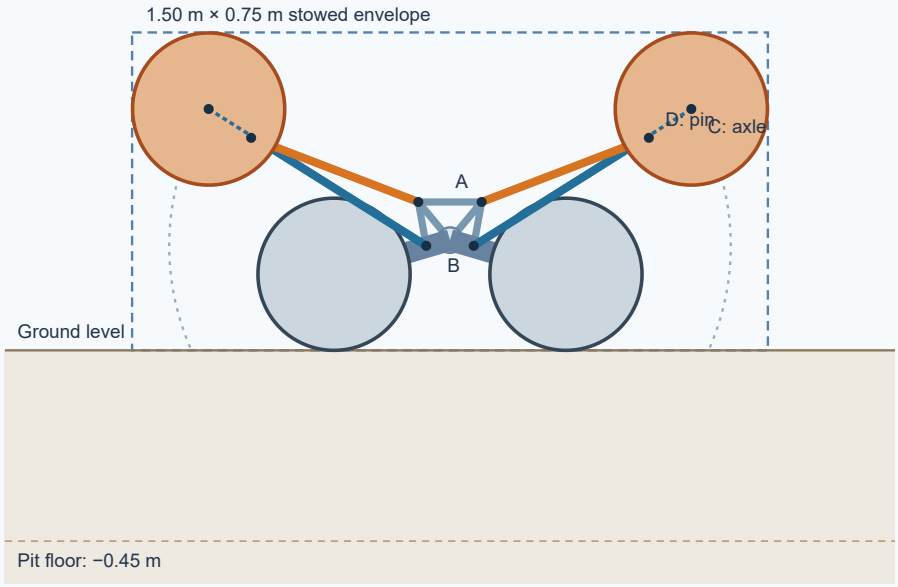
Four aluminum cross-tubes: upper/lower footprints can overlap. Dashed blue: 1500 x 750 stowed limit.

Plate outlines are schematic. Projection overlap does not prove collision. Heights and equipment are not shown.

Travel-pose plan view. Four wheels and four drums are shown at their model stations. Top-down SVG

Travel

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



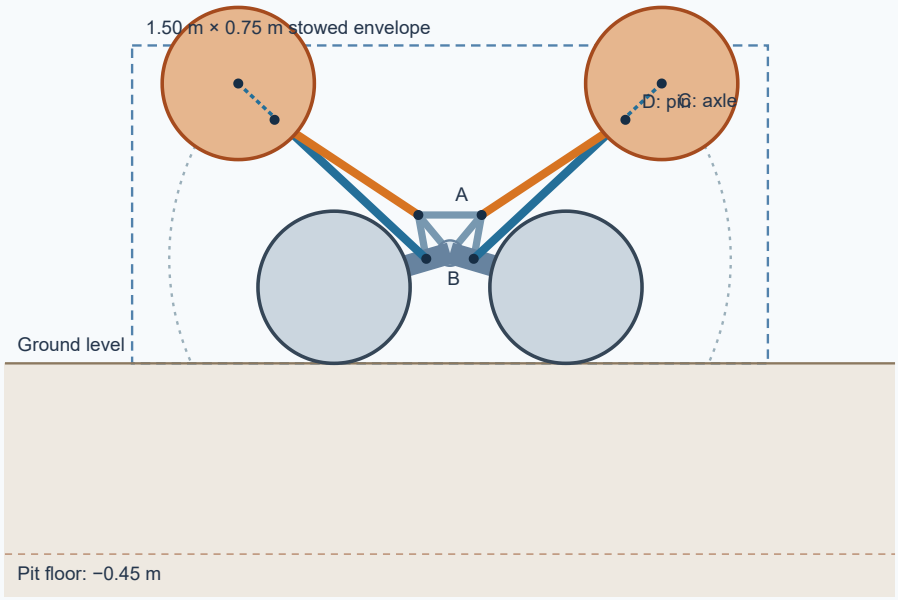
Actuator: 422.6 mm · Fixed arm: 606.5 mm · Drum Ø 359.0 mm

Drum top: +0.749 m · Drum bottom: +0.390 m · A: 0.350 m · B: 0.247 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -121.0 mm / top -4.2 mm · Ground is y = 0

Deposition · both pairs

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



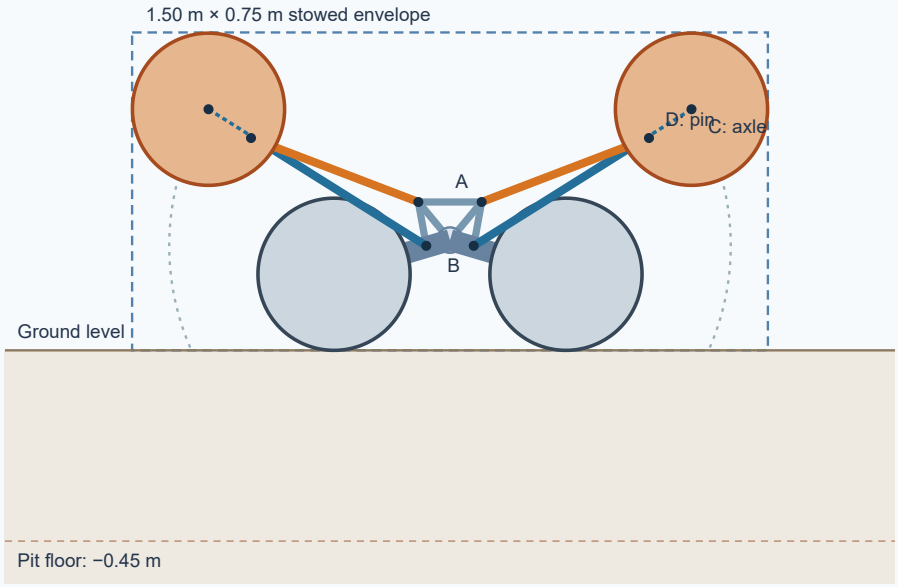
Actuator: 406.8 mm · Fixed arm: 606.5 mm · Drum Ø 359.0 mm

Drum top: +0.840 m · Drum bottom: +0.481 m · A: 0.350 m · B: 0.247 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -121.0 mm / top -4.2 mm · Ground is y = 0

Compact · starting envelope

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



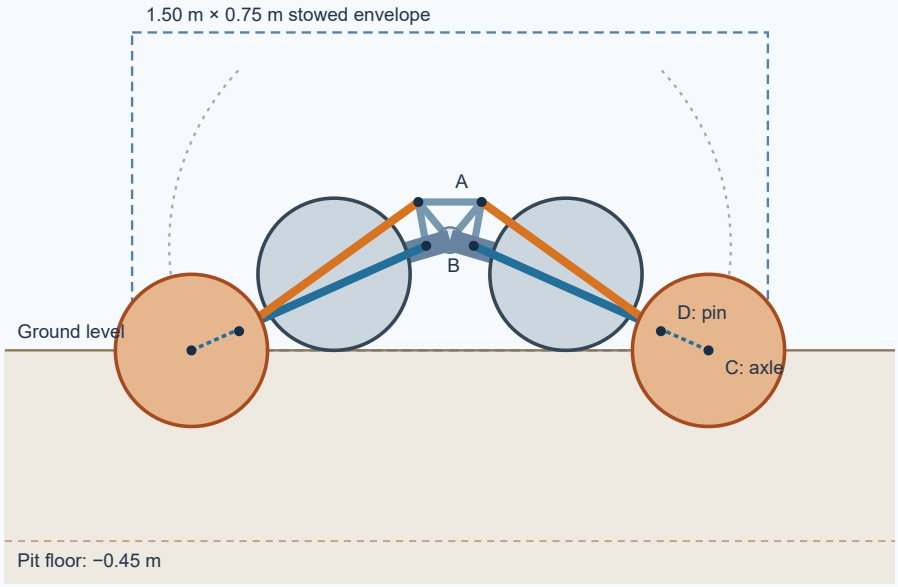
Actuator: 422.7 mm · Fixed arm: 606.5 mm · Drum Ø 359.0 mm

Drum top: +0.748 m · Drum bottom: +0.389 m · A: 0.350 m · B: 0.247 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -121.0 mm / top -4.2 mm · Ground is y = 0

Maximum extension · plunge / recovery reach

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



Actuator: 521.4 mm · Fixed arm: 606.5 mm · Drum Ø 359.0 mm

Drum top: +0.179 m · Drum bottom: -0.180 m · A: 0.350 m · B: 0.247 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -121.0 mm / top -4.2 mm · Ground is y = 0

Blue: arm · orange: actuator. Each visible drum represents a shaft-linked pair. Drawings are not CAD.

Travel and deposition mass properties

Pose / loading	Longitudinal CG	CG height	Yaw inertia
Travel · fully loaded	0.00 mm	471.2 mm	48.2 kg·m²
Deposition · fully loaded	0.00 mm	545.6 mm	39.2 kg·m²
Deposition turn · one pair full	273.3 mm	515.7 mm	23.0 kg·m²
Deposition · empty	5.12e-15 mm	413.4 mm	6.89 kg·m²

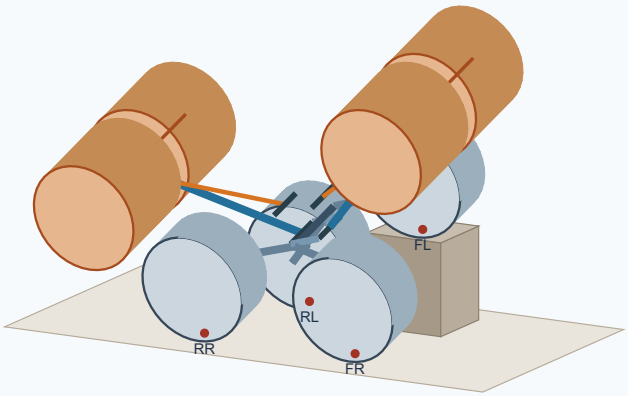
Added pose-change duration: 3.18 s. Added electrical energy: 1,068 J. Both pairs move together while full before dumping, then return empty after both dumps.

Raising reduces yaw inertia only when the drums move inward. Higher CG can reduce dynamic stability. One wheelbase must satisfy both travel and deposition checks. Half-loaded arm repositioning and dynamic dumping require controller and hardware validation.

One wheel on the rock / travel pose

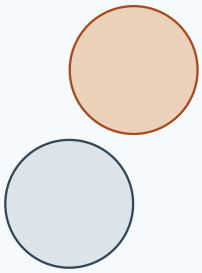
Front-left raised: 300.0 mm · chassis pitch 17.2° · roll 14.5°

Same kinematic pose as the model; arms held in travel position.



Common roll removed

Closest cylinder pair: drum 1 / FL



Radial circle gap: +58.03 mm

Axial end gap: -192.28 mm

Finite-cylinder gap: +58.03 mm

Required clearance: 15.00 mm

One centered arm/actuator per drum pair · blue: arms · orange: actuators · charcoal: aluminum cross-tubes · gray: wheels · tan: drums.

Support block shows the prescribed height; its shape and width are illustrative. Actual rock climbing is not simulated.

Contact model omits camber effects. The right section also needs the axial-width check; circles alone can over-reject.

Parallel-cylinder clearance is the default model. This section shows the closest pair for the illustrated front-left raise.

Plate-frame member schematic

Plate frame: provisional member topology

Lengths and directions follow the mounting coordinates; this is not the manufactured cutout outline.

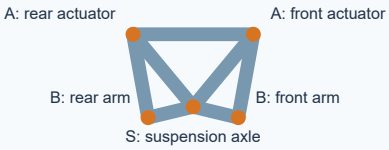


Plate thickness: 6.35 mm (fixed)

Common strip width: 17.07 mm

cross-tube ID / OD: 25.40 / 31.84 mm

Plate spacing / cross-tube span: 160.0 mm

Rocker bearing spacing: 288.9 mm

Wheel-to-rocker gap: 12.7 mm

Aluminum arm box: 50.24 x 53.45 mm

Arm wall: 1.4541 mm

Central support: pin, balanced mirrored gravity only.

cross-tube material properties are PROVISIONAL.

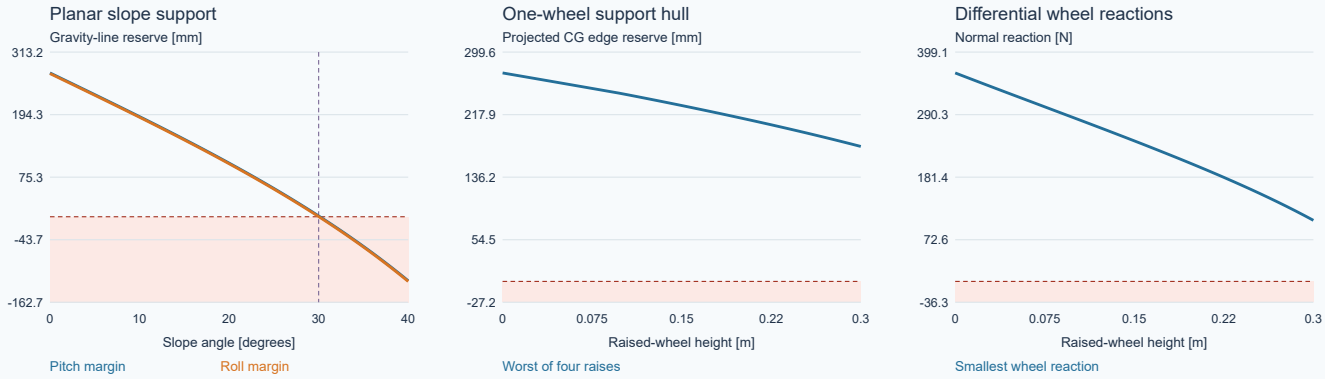
Spoke buckling is a strip screen,

not whole-plate buckling validation.

Static slope and articulation curves

Loaded travel pose: gravity stability screens

One wheel raised at a time on horizontal platforms. Curves are model calculations; no physical rock profile is simulated.



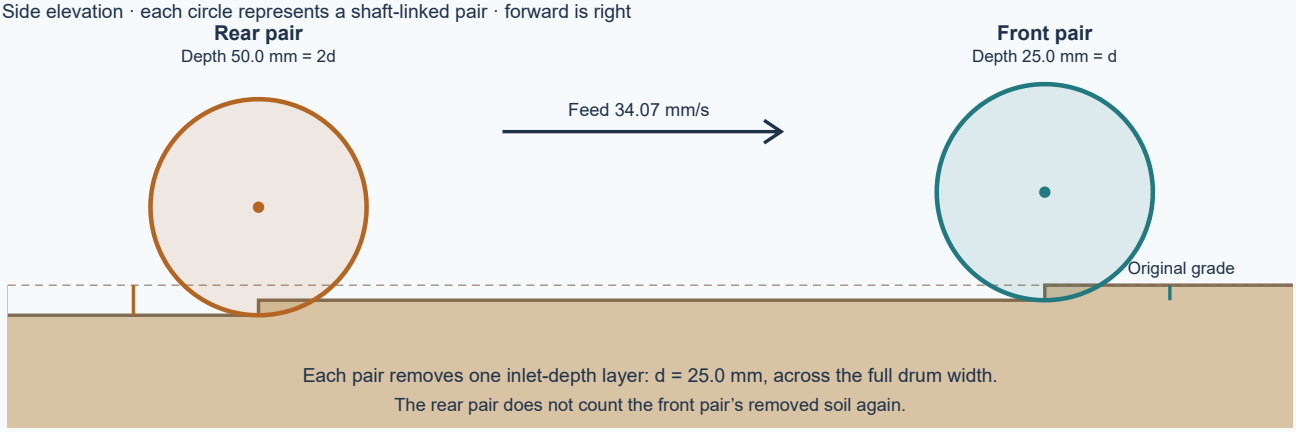
EXCAVATION / FORWARD FEED

Drive straight with the drums lowered

Each pair cuts one inlet-depth fresh layer across its full drum width. Radial protrusion equals inlet depth; the rear cuts a second layer in the same tracks. Forward advance per scoop remains limited by inlet depth.

Moving excavation · two fresh layers

Side elevation · each circle represents a shaft-linked pair · forward is right



Staged entry: front starts → rear reaches that start and lowers → front lifts when full → rear finishes.

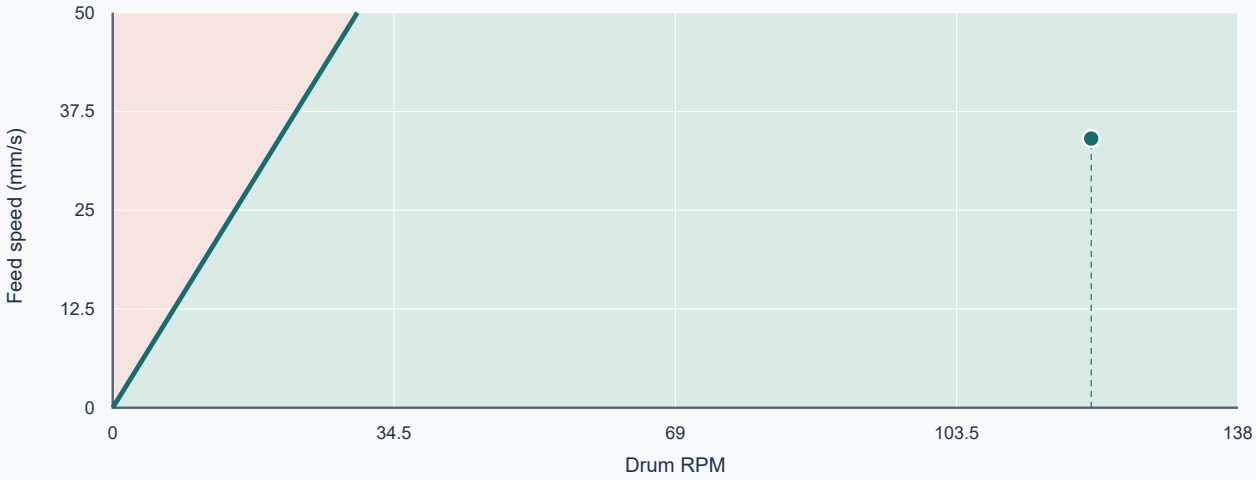
Productive lane: 3.115 m · total feed travel: 4.421 m · feed time: 130.0 s

The maximum-extension view is a separate plunge/recovery reach screen; it is not this operating pose.

Schematic soil boundary; entry/endcap soil is not credited. Soil flow and recovery driving require validation.

Feed speed and drum RPM

Forward advance per scoop ≤ inlet depth · an uncalibrated feed screen



Current: 120.00 rpm, 34.07 mm/s

Forward advance: 4.26 mm/scoop · inlet depth: 25.00 mm

Maximum feed = inlet_depth × scoops/revolution × RPM / 60; at this RPM: 200.00 mm/s.

Green: passes this feed screen. Red: excessive advance per scoop. Motor, reach and traction checks still apply.

This volume/bite constraint does not establish measured capture, cutting loads or high-speed soil retention.

Excavation quantity	Evaluated value
Inlet depth / fresh-layer cut depth	25.0 mm
Inlet width (full drum width)	344.6 mm
Front depth below original grade	25.0 mm
Rear depth below original grade	50.0 mm
Straight excavation feed speed	34.1 mm/s
Excavation drum RPM	120.0 rpm
Advance per scoop	4.26 mm
Feed limit at selected drum RPM	200.0 mm/s
Productive excavation lane length	3,115 mm
Feed travel including tandem staging	4,421 mm
Straight-feed duration	130.0 s
Feed-drive electrical energy	1,728 J
Feed-drive peak electrical power	18.5 W

Excavation quantity	Evaluated value
Plunge/recovery: axle reaches ground level	0.15 mm

Feed-drive electrical work and peak power are included in digging totals. The old stationary-plunge soil-volume calculation is inactive. Maximum extension remains a separate capability envelope. Reaching the ground with an axle does not verify motor clearance or driving on the drums.

02 / PREDICTED OUTPUT

Performance & mass

Complete-cycle estimates and continuous MDO estimates are labeled separately.

Prediction	Value	Meaning
Full cycle duration	272.5 s	Includes digging, actuation, travel, dumping and relocation; overlap removed once.
Energy per cycle	21.8 Wh	Electrical energy including auxiliaries.
Mission energy · complete cycles	65.9 Wh	Includes idle electronics for remaining mission time.
Mission energy · continuous estimate	71.9 Wh	Used by the existing battery constraint and MDO objective.
Peak electrical power	1,220 W	Largest modeled simultaneous demand; not an average power rating.
Loaded mass	148.0 kg	Dry mass + all four drums' regolith payload.
Traversal timing mode	calculated	BWR/motor-limited candidate profiles; legacy turn screens remain enforced.
Loaded outbound route	25.7 s	Includes route turns; excludes deposition and relocation.
Empty return route	25.7 s	Includes route turns; excludes deposition and relocation.
Actual route peak speed	1.00 m/s	Calculated profile, not measured vehicle performance.
Score · complete-cycle estimate	1,561.45 points	Uses complete deliveries and idle remainder; separate from the continuous MDO objective.
Empty rover · 0° climb	Slip 0.1668	BWR, GRC-1, equal mean wheel loads. Force shortfall per wheel: 0.00 N. Soil-fit range flags: 0. Acceptance: solved slip strictly below 1.
Full rover · 0° climb	Slip 0.1606	BWR, GRC-1, equal mean wheel loads. Force shortfall per wheel: 0.00 N. Soil-fit range flags: 2. Acceptance: solved slip strictly below 1.

Dry mass breakdown

Each assembly is counted once.

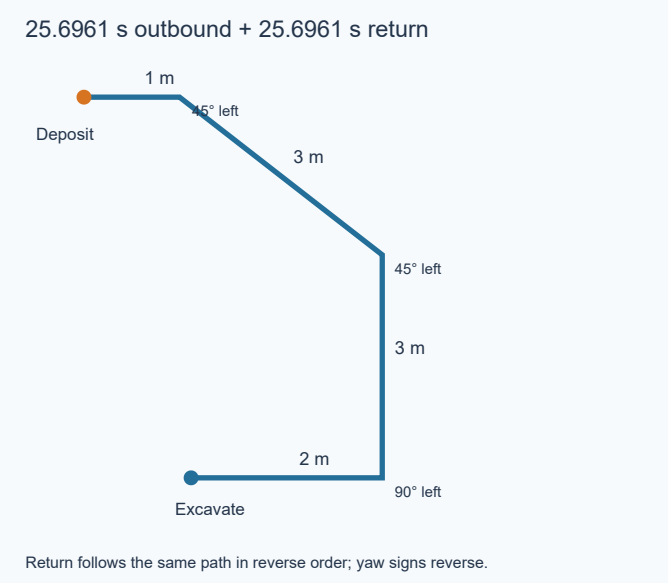
Equipment allowance	**6.00 kg**
Frame	**5.02 kg**
Four bare wheels	**6.49 kg**
Four drive assemblies	**4.15 kg**
Four drums + two drive assemblies	**14.0 kg**
Battery	**1.82 kg**
Arms + actuators	**5.84 kg**

Electrical energy per cycle

Work / provisional efficiency, plus auxiliaries.

Digging	**15.8 Wh**
Dumping	**0.14 Wh**
Travel + deposition turn	**1.73 Wh**
Excavation actuators	**1.86 Wh**
Deposition pose changes	**0.30 Wh**
Fresh-site move	**0.07 Wh**
Electronics	**1.89 Wh**

Route & cycle accounting



25.6961 s outbound + 25.6961 s return

Position each pair → drive the staged straight excavation pass → raise to travel → outbound → move to deposition pose → dump one pair → turn → dump second pair → return empty drums to travel pose → return route → relocate to a fresh lane.

Moving excavation has no lowering/cutting overlap credit. Full-range arm motion and load allowances conservatively screen the plunge/recovery sweep. The deposition turn and fresh-site relocation are additional to the timed route. Modeled full cycle: 272.5 s.

03 / GAPS TO RESOLVE

Requirements & model checks

Failures are listed first. “Not assessed” means the model does not verify that capability.

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Turning: traction reserve (excluded from optimizer) `turn_traction_margin`	Reserve: -261.1 N	Reserve ≥ 0	-261.1 N	Fail
Dry rover mass `dry_mass`	43.3 kg	≤ 80.0 kg	36.7 kg	Pass
Compact overall length `rover_length`	1,499 mm	≤ 1,500 mm	1.50 mm	Pass
Overall width `rover_width`	749.2 mm	≤ 750.0 mm	0.75 mm	Pass
Compact overall height `rover_height`	748.3 mm	≤ 750.0 mm	1.67 mm	Pass
Ground clearance: lower axle after wheel/bar deductions `ground_clearance_margin`	Reserve: 17.0 mm	Reserve ≥ 0	17.0 mm	Pass
Empty rover: 0° BWR climb `empty_slope_traversal_margin`	0.166831	Slip < 1 (strict)	0.83	Pass
Full rover: 0° BWR climb `loaded_slope_traversal_margin`	0.160636	Slip < 1 (strict)	0.84	Pass
Excavation shaft-power reserve `drum_power_margin`	Reserve: 604.6 W	Reserve ≥ 0	604.6 W	Pass
Excavation: penetration reserve `penetration_margin`	Reserve: 372.0 N	Reserve ≥ 0	372.0 N	Pass
Excavation: traction reserve (excluded from optimizer) `dig_traction_margin`	Reserve: 0.30 N	Reserve ≥ 0	0.30 N	Pass
Inlet depth: radial geometry reserve `cut_depth_margin`	Reserve: 19.9 mm	Reserve ≥ 0	19.9 mm	Pass
Ground contact pressure `ground_pressure_margin`	17.1 kPa	≤ 25.0 kPa	7.94 kPa	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Straight travel: motor-force reserve `straight_motor_margin`	Reserve: 510.4 N	Reserve ≥ 0	510.4 N	Pass
Turning: motor-force reserve `turn_motor_margin`	Reserve: 5,316 N	Reserve ≥ 0	5,316 N	Pass
Straight travel: traction reserve `straight_traction_margin`	Reserve: 177.1 N	Reserve ≥ 0	177.1 N	Pass
Straight travel: shaft-power reserve `straight_power_margin`	Reserve: 2,287 W	Reserve ≥ 0	2,287 W	Pass
Turning: shaft-power reserve `turn_power_margin`	Reserve: 2,537 W	Reserve ≥ 0	2,537 W	Pass
Battery usable energy `battery_energy_margin`	71.9 Wh	≤ 203.9 Wh	131.9 Wh	Pass
Battery peak-power capacity `battery_power_margin`	1,220 W	≤ 1,699 W	478.7 W	Pass
Time remaining after one complete cycle `complete_cycle_margin`	Reserve: 627.5 s	Reserve ≥ 0	627.5 s	Pass
Pit depth margin `pit_depth_margin`	Reserve: 270.3 mm	Reserve ≥ 0	270.3 mm	Pass
Transport clearance margin `transport_clearance_margin`	Reserve: 309.3 mm	Reserve ≥ 0	309.3 mm	Pass
Travel drum height above wheel-diameter step `wheel_first_margin`	Reserve: 30.9 mm	Reserve ≥ 0	30.9 mm	Pass
Deposition pose: triangle reach reserve `deposition_reach_margin`	Reserve: 11.4 mm	Reserve ≥ 0	11.4 mm	Pass
Deposition command below maximum extension `deposition_position_margin`	Reserve: 114.5 mm	Reserve ≥ 0	114.5 mm	Pass
Deposition drum above ground `deposition_clearance_margin`	Reserve: 400.7 mm	Reserve ≥ 0	400.7 mm	Pass
Startup position margin `startup_position_margin`	Reserve: 0.10 mm	Reserve ≥ 0	0.10 mm	Pass
Dig position margin `dig_position_margin`	Reserve: 98.7 mm	Reserve ≥ 0	98.7 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Pivot spacing margin `pivot_spacing_margin`	Reserve: 25.1 mm	Reserve ≥ 0	25.1 mm	Pass
Pivot mount margin `pivot_mount_margin`	Reserve: 359.0 mm	Reserve ≥ 0	359.0 mm	Pass
Pivot longitudinal margin `pivot_longitudinal_margin`	Reserve: 217.6 mm	Reserve ≥ 0	217.6 mm	Pass
Actuator pivot longitudinal margin `actuator_pivot_longitudinal_margin`	Reserve: 198.9 mm	Reserve ≥ 0	198.9 mm	Pass
Bottom mount margin `bottom_mount_margin`	Reserve: 67.1 mm	Reserve ≥ 0	67.1 mm	Pass
Stroke margin `stroke_margin`	Reserve: 58.8 mm	Reserve ≥ 0	58.8 mm	Pass
Triangle retracted margin `triangle_retracted_margin`	Reserve: 27.2 mm	Reserve ≥ 0	27.2 mm	Pass
Triangle extended margin `triangle_extended_margin`	Reserve: 54.1 mm	Reserve ≥ 0	54.1 mm	Pass
Toggle margin `toggle_margin`	Reserve: 428.6 mm	Reserve ≥ 0	428.6 mm	Pass
Actuator toggle margin `actuator_toggle_margin`	Reserve: 278.9 mm	Reserve ≥ 0	278.9 mm	Pass
Wheel clearance margin `wheel_clearance_margin`	Reserve: 0.30 mm	Reserve ≥ 0	0.30 mm	Pass
Deployed height margin `deployed_height_margin`	Reserve: 1,660 mm	Reserve ≥ 0	1,660 mm	Pass
Actuator force capacity `actuator_force_margin`	3,999 N	≤ 4,003 N	3.98 N	Pass
Excavation torque reserve `drum_torque_margin`	Reserve: 41.6 N·m	Reserve ≥ 0	41.6 N·m	Pass
Dumping torque reserve `dump_torque_margin`	Reserve: 74.0 N·m	Reserve ≥ 0	74.0 N·m	Pass
Minimum payload reserve `payload_margin`	Reserve: 104.6 kg	Reserve ≥ 0	104.6 kg	Pass
Excavated trench diameter reserve `trench_traverse_margin`	Reserve: 309.0 mm	Reserve ≥ 0	309.0 mm	Pass
Fresh site margin `fresh_site_margin`	Reserve: 50.8 mm	Reserve ≥ 0	50.8 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Frame section margin `frame_section_margin`	Reserve: 7.41 mm	Reserve ≥ 0	7.41 mm	Pass
Arm section margin `arm_section_margin`	Reserve: 11.1 mm	Reserve ≥ 0	11.1 mm	Pass
Structural stress envelope `structure_stress_margin`	77.9 MPa	≤ 120.0 MPa	42.1 MPa	Pass
Structural deflection envelope `structure_deflection_margin`	1.02 mm	≤ 5.00 mm	3.98 mm	Pass
Structural buckling reserve `structure_buckling_margin`	Reserve: 5,565 N	Reserve ≥ 0	5,565 N	Pass
Level-route dynamic tip reserve `mobility_stability_margin`	Reserve: 246.4 mm	Reserve ≥ 0	246.4 mm	Pass
Asymmetric dump tip reserve `tip_stability_margin`	Reserve: 0.30 mm	Reserve ≥ 0	0.30 mm	Pass
Rock pose: geometric reach reserve `obstacle_pose_margin`	Reserve: 215.7 mm	Reserve ≥ 0	215.7 mm	Pass
Rock endpoint: global tip reserve `obstacle_tip_margin`	Reserve: 176.3 mm	Reserve ≥ 0	176.3 mm	Pass
Obstacle drum–wheel clearance `obstacle_drum_wheel_margin`	58.0 mm	≥ 15.0 mm	43.0 mm	Pass
30° slope: static tip reserve `slope_tip_margin`	Reserve: 0.30 mm	Reserve ≥ 0	0.30 mm	Pass
Rock path: global tip reserve `rock_sweep_tip_margin`	Reserve: 176.3 mm	Reserve ≥ 0	176.3 mm	Pass
Rock path: smallest wheel reaction `rock_sweep_wheel_load_margin`	Reserve: 106.3 N	Reserve ≥ 0	106.3 N	Pass
Available excavation area `excavation_area_margin`	Reserve: 0.17 m^2	Reserve ≥ 0	0.17 m^2	Pass
Fresh-site move: drive/traction reserve `site_drive_margin`	Reserve: 162.9 N	Reserve ≥ 0	162.9 N	Pass
Plate geometry reserve `plate_geometry_margin`	Reserve: 0.10 mm	Reserve ≥ 0	0.10 mm	Pass
Frame packaging reserve `frame_packing_margin`	Reserve: 51.3 mm	Reserve ≥ 0	51.3 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Plate strip stress `plate_stress_margin`	29.8 MPa	≤ 120.0 MPa	90.2 MPa	Pass
Plate strip buckling reserve `plate_buckling_margin`	Reserve: 5,565 N	Reserve ≥ 0	5,565 N	Pass
aluminum cross-tube bending stress `cross_tube_bending_margin`	119.9 MPa	≤ 120.0 MPa	0.10 MPa	Pass
Actuator catalog stroke domain `actuator_stroke_domain_margin`	Reserve: 111.7 mm	Reserve ≥ 0	111.7 mm	Pass
Actuator load-curve domain `actuator_curve_margin`	Reserve: 3.98 N	Reserve ≥ 0	3.98 N	Pass
Forward advance per scoop: inlet-depth reserve `feed_margin`	4.26 mm	≤ 25.0 mm	20.7 mm	Pass
Moving excavation: operating-pose reach `operating_reach_margin`	Reserve: 22.2 mm	Reserve ≥ 0	22.2 mm	Pass
Plunge/recovery: axle reaches ground level `axle_ground_margin`	-0.15 mm	≤ 0.00 mm	0.15 mm	Pass
Moving excavation: geometry domain `excavation_domain_margin`	Reserve: 154.5 mm	Reserve ≥ 0	154.5 mm	Pass
Excavation feed: motor-force reserve `feed_motor_margin`	Reserve: 6,939 N	Reserve ≥ 0	6,939 N	Pass
Excavation feed: traction reserve `feed_traction_margin`	Reserve: 0.30 N	Reserve ≥ 0	0.30 N	Pass
Excavation feed: shaft-power reserve `feed_power_margin`	Reserve: 2,666 W	Reserve ≥ 0	2,666 W	Pass
Lin act side arm offset search range `lin_act_side_arm_offset_bounds`	-121.0 mm	-300.0 mm to -50.8 mm	70.2 mm	Pass
Lin act top arm offset search range `lin_act_top_arm_offset_bounds`	-4.24 mm	-150.0 mm to 150.0 mm	145.8 mm	Pass
Width of each drum search range `drum_length_bounds`	344.6 mm	120.0 mm to 359.0 mm	14.4 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Excavation drum RPM search range `drum_rpm_bounds`	120.0 rpm	5.00 rpm to 120.0 rpm	0.00 rpm	Pass
Straight excavation feed speed search range `feed_speed_bounds`	0.03 m/s	0.0001 m/s to 0.50 m/s	0.03 m/s	Pass
Dumping speed search range `dump_rpm_bounds`	60.0 rpm	5.00 rpm to 60.0 rpm	0.00 rpm	Pass
Drum reduction ratio search range `drum_gear_ratio_bounds`	40.3	9.00 to 125.0	31.3	Pass
Inlet depth / fresh-layer cut depth search range `inlet_depth_bounds`	25.0 mm	3.00 mm to 25.0 mm	0.00 mm	Pass
Arm wall thickness search range `arm_wall_bounds`	1.45 mm	1.00 mm to 5.00 mm	0.45 mm	Pass
Battery capacity search range `battery_capacity_bounds`	254.8 Wh	10.0 Wh to 300.0 Wh	45.2 Wh	Pass
Travel actuator length search range `travel_actuator_length_bounds`	422.6 mm	403.1 mm to 606.3 mm	19.5 mm	Pass
Deposition actuator length search range `deposition_actuator_length_bounds`	406.8 mm	403.1 mm to 606.3 mm	3.75 mm	Pass
Compact actuator length search range `actuator_retracted_length_bounds`	422.7 mm	403.1 mm to 606.3 mm	19.6 mm	Pass
Maximum-extension actuator length search range `actuator_extended_length_bounds`	521.4 mm	403.1 mm to 606.3 mm	84.9 mm	Pass
Actuator pivot height search range `top_pivot_height_bounds`	350.0 mm	350.0 mm to 700.0 mm	0.00 mm	Pass
Actuator pivot longitudinal position search range `actuator_pivot_x_bounds`	74.6 mm	20.0 mm to 300.0 mm	54.6 mm	Pass
Arm pivot height search range `bottom_pivot_height_bounds`	246.6 mm	90.0 mm to 300.0 mm	53.4 mm	Pass
Arm length search range `arm_length_bounds`	606.5 mm	300.0 mm to 700.0 mm	93.5 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Arm pivot longitudinal position search range `pivot_x_bounds`	56.0 mm	50.0 mm to 300.0 mm	5.96 mm	Pass
Wheelbase search range `wheelbase_bounds`	547.2 mm	380.0 mm to 650.0 mm	102.8 mm	Pass
Wheel-center track search range `track_width_bounds`	544.7 mm	500.0 mm to 699.0 mm	44.7 mm	Pass
Aluminum arm box width search range `arm_box_width_bounds`	50.2 mm	25.0 mm to 100.0 mm	25.2 mm	Pass
Aluminum arm box height search range `arm_box_height_bounds`	53.5 mm	25.0 mm to 100.0 mm	28.5 mm	Pass
aluminum cross-tube span / plate spacing search range `cross_tube_span_bounds`	160.0 mm	160.0 mm to 460.0 mm	0.00 mm	Pass
Aluminum cross-tube wall thickness search range `cross_tube_wall_bounds`	3.22 mm	1.00 mm to 6.00 mm	2.22 mm	Pass
Plate strip width search range `plate_spoke_width_bounds`	17.1 mm	12.0 mm to 70.0 mm	5.07 mm	Pass
Rocker box width search range `rocker_box_width_bounds`	50.4 mm	20.0 mm to 65.0 mm	14.6 mm	Pass
Rocker box height search range `rocker_box_height_bounds`	51.3 mm	25.0 mm to 90.0 mm	26.3 mm	Pass
Rocker wall search range `rocker_wall_bounds`	1.52 mm	1.00 mm to 5.00 mm	0.52 mm	Pass
Axle outer diameter search range `axle_outer_diameter_bounds`	33.7 mm	25.0 mm to 90.0 mm	8.73 mm	Pass
Axle wall search range `axle_wall_bounds`	1.02 mm	1.00 mm to 6.00 mm	0.02 mm	Pass
Rocker pivot above wheel centers search range `rocker_pivot_offset_bounds`	80.6 mm	20.0 mm to 150.0 mm	60.6 mm	Pass
Actuator catalog stroke (continuous) search range `actuator_catalog_stroke_bounds`	162.5 mm	50.8 mm to 1,016 mm	111.7 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Drum outer diameter search range `drum_diameter_bounds`	359.0 mm	160.0 mm to 359.0 mm	0.00 mm	Pass
Wheel diameter search range `wheel_diameter_bounds`	359.0 mm	180.0 mm to 359.0 mm	0.00 mm	Pass
Wheel width search range `wheel_width_bounds`	180.0 mm	50.0 mm to 359.0 mm	130.0 mm	Pass
Drum-supported recovery driving	Axle-to-ground reach is screened geometrically. Drum traction, motor-ground clearance, recovery loads and actual escape capability require hardware and soil tests.	—	—	Not assessed
Slope motor limits / sideways sliding	The separate BWR climb checks accept slip<1 only. Motor capability on slopes, lateral sliding and controller behavior are not evaluated.	—	—	Not assessed
Real 30 cm rock traversal	Static poses and clearance screens do not establish the driving forces or actual terrain-contact sequence.	—	—	Not assessed
50 cm ditch crossing	Ditch width/radius and contact sequence have not been defined.	—	—	Not assessed
Continuous motor operation	Stator-current limits and thermal duty remain undecided.	—	—	Not assessed
Connections & material validation	Bonded inserts, bushings, bolts, local failure and selected aluminum cross-tube properties need separate checks.	—	—	Not assessed
Soil and skid-steer predictions	Rolling, traction, capture and scrub relationships remain uncalibrated.	—	—	Not assessed

Positive margins satisfy the modeled criterion. Status uses the saved model's numerical tolerance. Search ranges are hidden by default and available in the filter; they remain part of saved feasibility. Margins use their own units and cannot be ranked directly across unlike checks.

04 / ENGINEERING LOADS

Loads & structural response

Values identify their load case and whether they apply per component or to the whole rover.

Frame gravity solution is valid for its declared load case. Structural load factor: 1.5; structural safety factor: 2.

Quantity / scope	Prediction	Load case / interpretation
Required force per actuator `required_actuator_force`	3,999 N	Loaded raising/extraction envelope, including the model's breakout allowance.
Frame gravity reaction at each actuator `frame_gravity_actuator_force`	5,122 N	Mirrored loaded gravity sweep; includes the structural load factor. Separate from actuator sizing load.
Nominal force per wheel `frame_nominal_wheel_reaction`	362.8 N	Equal-share level-ground weight. Not a rock or slope wheel-load envelope.
Total cutting force `cut_force`	105.7 N	Across four drums at the selected bite and engagement.
Worst modeled contact pressure `ground_pressure`	17.1 kPa	Route pressure estimate, including load transfer and rocker factor.
Structural stress envelope `structure_stress`	77.9 MPa	Envelope of local member stresses; not one assembled-frame FEA result.
Structural deflection envelope `structure_deflection`	1.02 mm	Conservative combination of local/relative deflections.
Upper/lower mount relative displacement `frame_mount_relative_deflection`	0.50 mm	Frame gravity sweep.
Rocker-end flexure `rocker_deflection`	0.44 mm	Local beam-screen displacement.
Suspension-axle flexure `axle_deflection`	0.09 mm	Axle-overhang screen.
Arm-tip flexure `arm_deflection`	1.02 mm	Relative arm beam displacement.
Loaded CG height `loaded_cg_z`	471.2 mm	Fully loaded travel pose, above level ground.

05 / SCOPE & TRACEABILITY

Model notes & engineering data

Keep assumptions and validity limits attached to the predictions.

Obstacle and structural interpretation

Requested endpoint rock pose is geometrically reachable; this alone does not establish traversal.

Obstacle drum–wheel clearance

Requested-height obstacle clearance screen. Parallel finite cylinders: shared roll is removed, then circular-plane and axial-width gaps are combined. Negative gap means overlap of the ideal solids. A common roll rotation preserves relative clearance in both models; rocker articulation still changes relative geometry.

Ditch requirement remains unverified. Static slope tipping assumes adequate friction; it does not verify climbing. The 5 mm structural limit applies to the modeled envelope of local deflections, not one assembled-frame displacement.

For the plate frame, aluminum cross-tube stiffness/strength are PROVISIONAL. Connections, dynamic obstacle loads, cutting/recovery frame cases and whole-plate buckling need separate verification.

Standard X44 motors · assumptions

Four wheel motors at 100:1; two drum motors at 40.30:1 (exploratory gearing).

Electrical energy = output-shaft mechanical work / combined efficiency. Drive: 72%; drums: 70%. Both efficiencies are provisional.

Continuous operation unverified. Current-limit inputs of zero mean undecided. Shaft torque-speed capability is enforced separately from energy accounting.

PA-04 actuators · 900 lbf / 12 V

Two actuators. Continuous catalog stroke: 162.5 mm; installed hardware limits: 523.7 mm to 686.2 mm. These include 161.3 mm added to each endpoint. Compact, travel, deposition and digging commands are compared with these limits diagnostically; packaging is not an active constraint.

Mass per actuator: 2.44 kg. Raising envelope: 0.00991 m/s and 14.0 A; raising time: 9.97 s, lowering time: 9.97 s. Lowering assumes rated-load speed and 14 A. Electrical energy uses 12 V × current × time for both actuators, including motor losses.

Mass fit supports 2–40 inch strokes; load curves support 0–900 lbf. Guarded out-of-domain diagnostics retain failed checks. Independent hardware, gravity-assisted lowering, duty cycle, temperature and installation-hardware mass remain unverified. A continuous stroke must be replaced with an available catalog size and reevaluated before purchase.

Removed screens: diagnostics only

- `actuator_body_margin`: User removed this constraint; the reserve is diagnostic only and does not affect optimization or feasibility. Actual actuator packaging remains unverified.
- `dig_froude_margin`: User removed the uncalibrated speed cap; constant capture/lift response assumes no additional high-speed loss.
- `dump_froude_margin`: User removed the uncalibrated speed cap; actual dumping/retention speed response remains unverified.
- `chassis_clearance_margin`: Legacy filled rectangular envelope does not represent the open plate frame; actual frame/equipment interference remains unverified.
- `dig_reach_margin`: Compatibility alias of operating_reach_margin; use the active operating_reach_margin and axle_ground_margin checks.

06 / MDO ONLY

Optimization study

The candidate is separate from a released design baseline. Optimizer status: **True**. Saved status: failed_or_infeasible.

Continuous-score objective: 297.48 → 1,692.78 points. Complete-cycle estimate: 1,561.45 points.

Changed input	Starting value	Candidate value
Lin act side arm offset	-200.8 mm	-121.0 mm
Lin act top arm offset	50.8 mm	-4.24 mm
Width of each drum	340.0 mm	344.6 mm
Straight excavation feed speed	5.00 mm/s	34.1 mm/s
Drum reduction ratio	40.0	40.3
Inlet depth / fresh-layer cut depth	14.0 mm	25.0 mm
Arm wall thickness	1.59 mm	1.45 mm
Battery capacity	259.2 Wh	254.8 Wh
Travel actuator length	403.1 mm	422.6 mm
Deposition actuator length	403.1 mm	406.8 mm
Compact actuator length	403.1 mm	422.7 mm
Maximum-extension actuator length	606.0 mm	521.4 mm
Actuator pivot height	520.0 mm	350.0 mm
Actuator pivot longitudinal position	38.1 mm	74.6 mm
Arm pivot height	280.0 mm	246.6 mm
Arm length	558.8 mm	606.5 mm
Arm pivot longitudinal position	139.7 mm	56.0 mm
Wheelbase	600.0 mm	547.2 mm
Wheel-center track	570.0 mm	544.7 mm
Aluminum arm box width	50.8 mm	50.2 mm
Aluminum arm box height	50.8 mm	53.5 mm
Aluminum cross-tube wall thickness	2.00 mm	3.22 mm
Plate strip width	18.8 mm	17.1 mm
Rocker box width	50.8 mm	50.4 mm
Rocker box height	50.8 mm	51.3 mm
Rocker wall	1.59 mm	1.52 mm
Axle outer diameter	33.0 mm	33.7 mm
Axle wall	1.00 mm	1.02 mm
Rocker pivot above wheel centers	76.2 mm	80.6 mm
Actuator catalog stroke (continuous)	203.2 mm	162.5 mm

APPENDIX / NATIVE UNITS

Design inputs & output data

142 inputs supplied by file; 3 CLI overrides; 4 remaining inputs use model defaults. Optimizer-selected design values appear in the bounds table. Frozen design variables (held fixed): drum_diameter, wheel_diameter, wheel_width. Values below retain their native model units.

All effective inputs & model assumptions

Includes mission, soil, material properties and inactive legacy inputs. Source SHA-256: `c28db0377fcab51d92d133cfd151fe31374008aada62758970aefcfbe568bd10`.

Input port	Effective value	Native units
lin_act_side_arm_offset	-0.12101635	m
lin_act_top_arm_offset	-0.0042419375	m
drum_length	0.344625	m
drum_rpm	120	rpm
feed_speed	0.034071809	m/s
dump_rpm	60	rpm
drum_gear_ratio	40.297686	—
inlet_depth	0.025	m
arm_wall	0.0014540978	m
battery_capacity	254.82138	W*h
travel_actuator_length	0.42259925	m
deposition_actuator_length	0.40684987	m
actuator_retracted_length	0.42269925	m
actuator_extended_length	0.5213868	m
top_pivot_height	0.35	m
actuator_pivot_x	0.074648503	m
bottom_pivot_height	0.24662215	m
arm_length	0.60646036	m
pivot_x	0.055962804	m
wheelbase	0.54715025	m
track_width	0.54469233	m
arm_box_width	0.050238737	m
arm_box_height	0.053452265	m
cross_tube_span	0.16	m
cross_tube_wall	0.0032222047	m
plate_spoke_width	0.017073068	m
rocker_box_width	0.050381554	m
rocker_box_height	0.051252773	m
rocker_wall	0.0015198209	m
axle_outer_diameter	0.033730658	m
axle_wall	0.0010183709	m
rocker_pivot_offset	0.080582196	m

Input port	Effective value	Native units
actuator_catalog_stroke	0.16245683	m
minimum_ground_clearance	0.1524	m
clearance_radius_deflection_fraction	0.2	—
differential_bar_clearance_allowance	0.0254	m
empty_slope_degrees	0	deg
loaded_slope_degrees	0	deg
gravity	9.80665	m/s**2
run_time	900	s
traversal_time_one_way	45	s
travel_speed	1	m/s
acceleration	0.4	m/s**2
turn_angle	3.1415927	rad
overhead_time	8	s
electronics_power	25	W
actuator_idle_power	2	W
soil_density	1500	kg/m**3
berm_density	1500	kg/m**3
fill_fraction	0.9	—
delivery_efficiency	0.9	—
capture_fraction	0.65	—
cut_pressure	18000	Pa
internal_lift_factor	0.15	—
vertical_cut_fraction	0.5	—
force_imbalance	0.15	—
dump_revolutions	3	—
drum_efficiency	0.7	—
drum_gear_efficiency	0.8	—
drive_efficiency	0.72	—
drive_gear_efficiency	0.8	—
motor_rated_rpm	3000	rpm
traction_coefficient	0.6	—
rocker_load_factor	1.2	—
wheel_deflection_fraction	0.07	—
rolling_base	0.04	—
rolling_pressure_scale	200000	Pa
allowable_ground_pressure	25000	Pa
scrub_coefficient	0.25	—
turn_speed_fraction	0.35	—
turn_accel_fraction	0.5	—
chassis_height	0.35	m
drum_pair_gap	0.06	m
fixed_mass	6	kg
wheel_areal_mass	4	kg/m**2
drum_areal_mass	5	kg/m**2
motor_mass_per_watt	0.006	kg/W
gear_mass_per_torque	0.05	kg/(N*m)

Input port	Effective value	Native units
drive_fixed_mass	0.35	kg
drum_fixed_mass	0.5	kg
battery_specific_energy	140	W*h/kg
battery_usable_fraction	0.8	—
battery_c_rate	6.667	1/h
mass_score_weight	1	—
energy_score_weight	1	—
transport_clearance	0.08	m
mechanism_clearance	0.015	m
minimum_pivot_spacing	0.08	m
minimum_stroke	0.04	m
actuator_housing_allowance	0.04	m
pit_depth	0.45	m
actuator_efficiency	0.65	—
actuator_safety_factor	1	—
breakout_force	100	N
actuator_fixed_mass	0.6	kg
actuator_mass_per_force	0.0004	kg/N
actuator_mass_per_power	0.003	kg/W
actuator_mass_per_length	1	kg/m
stored_soil_density	1250	kg/m**3
usable_volume_fraction	0.9	—
scoops_per_revolution	4	—
max_dig_froude	0.3	—
max_dump_froude	0.2	—
drum_spinup_time	2	s
drum_torque_service_factor	1.25	—
minimum_dig_depth	0.005	m
minimum_payload	0.05	kg
fresh_site_distance	0.8	m
site_speed	0.1	m/s
available_excavation_area	12	m**2
tube_density	2700	kg/m**3
tube_youngs_modulus	6.9e+10	Pa
tube_yield_strength	2.4e+08	Pa
structure_safety_factor	2	—
structure_load_factor	1.5	—
allowable_deflection	0.005	m
joint_mass	2	kg
differential_mass	1	kg
obstacle_height	0.3	m
slope_angle	0.52359878	rad
frame_tube_diameter	0.055	m
frame_wall	0.002	m
arm_tube_diameter	0.05	m
plate_thickness	0.00635	m

Input port	Effective value	Native units
cross_tube_inner_diameter	0.0254	m
wheel_rocker_gap	0.0127	m
plate_hub_radius	0.03	m
plate_cutout_clearance	0.005	m
frame_mount_clearance	0.003	m
minimum_axle_overhang	0.01	m
plate_buckling_k	1	—
cross_tube_density	2700	kg/m**3
cross_tube_youngs_modulus	6.9e+10	Pa
cross_tube_yield_strength	2.4e+08	Pa
drive_motor_power	835.4	W
drum_motor_power	835.4	W
drive_gear_ratio	100	—
motor_bus_voltage	12	V
x44_motor_mass	0.34019428	kg
drive_gearbox_mass	0.697	kg
drum_gearbox_mass	0.753	kg
drive_stator_current_limit	0	A
drum_stator_current_limit	0	A
actuator_force_rating	4003.3995	N
actuator_speed	0.009906	m/s
actuator_installation_allowance	0.1613408	m
drum_diameter	0.359	m
wheel_diameter	0.359	m
wheel_width	0.18	m

Design variables and bounds

Optimizer search ranges and evaluated design variables. These ranges guide MDO and input validation; they are not all hardware requirements.

Port	Lower	Evaluated	Upper	Native units
lin_act_side_arm_offset	-0.3	-0.121016	-0.0508	m
lin_act_top_arm_offset	-0.15	-0.00424194	0.15	m
drum_length	0.12	0.344625	0.359	m
drum_rpm	5	120	120	rpm
feed_speed	0.0001	0.0340718	0.5	m/s
dump_rpm	5	60	60	rpm
drum_gear_ratio	9	40.2977	125	—
inlet_depth	0.003	0.025	0.025	m
arm_wall	0.001	0.0014541	0.005	m
battery_capacity	10	254.821	300	W*h
travel_actuator_length	0.403098	0.422599	0.606298	m
deposition_actuator_length	0.403098	0.40685	0.606298	m
actuator_retracted_length	0.403098	0.422699	0.606298	m
actuator_extended_length	0.403098	0.521387	0.606298	m
top_pivot_height	0.35	0.35	0.7	m
actuator_pivot_x	0.02	0.0746485	0.3	m
bottom_pivot_height	0.09	0.246622	0.3	m
arm_length	0.3	0.60646	0.7	m
pivot_x	0.05	0.0559628	0.3	m
wheelbase	0.38	0.54715	0.65	m
track_width	0.5	0.544692	0.699	m
arm_box_width	0.025	0.0502387	0.1	m
arm_box_height	0.025	0.0534523	0.1	m
cross_tube_span	0.16	0.16	0.46	m
cross_tube_wall	0.001	0.0032222	0.006	m
plate_spoke_width	0.012	0.0170731	0.07	m
rocker_box_width	0.02	0.0503816	0.065	m
rocker_box_height	0.025	0.0512528	0.09	m
rocker_wall	0.001	0.00151982	0.005	m
axle_outer_diameter	0.025	0.0337307	0.09	m
axle_wall	0.001	0.00101837	0.006	m
rocker_pivot_offset	0.02	0.0805822	0.15	m
actuator_catalog_stroke	0.0508	0.162457	1.016	m
drum_diameter (fixed)	0.16	0.359	0.359	m
wheel_diameter (fixed)	0.18	0.359	0.359	m
wheel_width (fixed)	0.05	0.18	0.359	m

All output ports

Output port	Value	Native units
dry_mass	43.32624	kg
payload_mass	104.66673	kg
wheel_mass	6.4872883	kg
drum_mass	14.008885	kg
drive_mass	4.1487771	kg
battery_mass	1.8201527	kg
rover_length	1.4985	m
rover_width	0.74925	m
rover_height	0.74832843	m
ground_pressure	17059.837	Pa
actual_travel_speed	1	m/s
dig_time	138.02223	s
travel_time	51.392292	s
dump_time	10	s
turn_time	4.1945735	s
cycle_time	272.48047	s
cycle_energy	21.77348	W*h
cycles	3.3029891	—
berm_volume	0.20742783	m**3
run_energy	71.917566	W*h
average_power	287.67026	W
peak_power	1220.2152	W
mass_points	1404.3567	—
energy_points	288.42443	—
construction_points	1692.7812	—
no_load_speed	1.4582122	m/s
loaded_motor_speed_limit	1.0419129	m/s
loaded_power_speed_limit	6.6819569	m/s
stall_drive_force	7333.0043	N
actuation_time	39.885921	s
lowering_time	9.9724964	s
arm_energy	6698.4567	J
arm_peak_power	336	W
deposition_reach_margin	0.011440332	m
deposition_position_margin	0.11453694	m
deposition_clearance_margin	0.40070901	m
required_actuator_force	3999.4211	N
linkage_mass	5.8441536	kg
stroke	0.098787549	m
stowed_x	0.56975	m
stowed_y	0.56882843	m
travel_x	0.5693911	m
travel_y	0.56940002	m

Output port	Value	Native units
deposition_x	0.49951672	m
deposition_y	0.66020901	m
extended_x	0.60994627	m
extended_y	-0.00015000003	m
sweep_x_min	0.49951672	m
sweep_x_max	0.66242317	m
actuator_leverage_max	5.8448993	—
actuator_attachment_distance	0.48546255	m
arm_physical_length	0.60646036	m
arm_extension_mass	0	kg
actuator_body_margin	-0.11684577	m
chassis_clearance_margin	0.027008695	m
stowed_length	1.4985	m
stowed_height	0.74832843	m
deployed_length	1.6838319	m
deployed_height	0.83970901	m
route_distance	18	m
route_turn_angle	360	deg
route_translation_time	37.268827	s
route_turn_time	14.123465	s
route_peak_acceleration	0.4	m/s**2
deposition_actuation_time	3.1794432	s
deposition_arm_energy	1067.9138	J
deposition_arm_peak_power	336	W
payload_volume	0.083733381	m**3
fresh_volume	0.10735049	m**3
storage_capacity	0.083733381	m**3
reachable_soil_mass	104.66673	kg
target_payload_mass	104.66673	kg
trench_depth	0.05	m
excavation_footprint	3.5817042	m**2
cut_force	105.67798	N
drum_froude	2.890437	—
dig_overlap_time	0	s
dig_power	859.21516	W
dump_power	126.03289	W
dig_energy	56827.805	J
dump_energy	499.92432	J
drum_power_margin	604.55196	W
drum_torque_margin	41.634924	N*m
dump_torque_margin	74.003669	N*m
dig_froude_margin	-2.590437	—
dump_froude_margin	-0.52260925	—
penetration_margin	372.04629	N
dig_traction_margin	0.3	N
cut_depth_margin	0.019875	m

Output port	Value	Native units
dig_reach_margin	0.02220294	m
payload_margin	104.61673	kg
structure_mass	5.0169834	kg
rocker_mass	0.92308847	kg
arm_linear_mass	0.7913595	kg/m
arm_envelope_radius	0.036677907	m
frame_section_margin	0.0074142937	m
arm_section_margin	0.011105586	m
structure_stress	77937897	Pa
structure_stress_margin	42062103	Pa
structure_deflection	0.0010223028	m
structure_deflection_margin	0.0039776972	m
structure_buckling_margin	5565.0381	N
empty_yaw_inertia	8.0601204	kg*m**2
empty_cg_x	1.024989e-17	m
empty_cg_z	0.3785069	m
loaded_yaw_inertia	48.156775	kg*m**2
loaded_cg_x	0	m
loaded_cg_z	0.47119778	m
half_yaw_inertia	28.108448	kg*m**2
half_cg_x	0.31150193	m
half_cg_z	0.45020696	m
loaded_sprung_mass	136.43381	kg
loaded_sprung_cg_x	0	m
loaded_sprung_cg_z	0.49563884	m
deposition_empty_yaw_inertia	6.8868515	kg*m**2
deposition_empty_cg_x	5.1249452e-18	m
deposition_empty_cg_z	0.41339852	m
deposition_loaded_yaw_inertia	39.166025	kg*m**2
deposition_loaded_cg_x	0	m
deposition_loaded_cg_z	0.54563647	m
deposition_half_yaw_inertia	23.026438	kg*m**2
deposition_half_cg_x	0.27327513	m
deposition_half_cg_z	0.5156898	m
relocation_time	17.806007	s
relocation_energy	246.73624	J
relocation_peak_power	20.788777	W
fresh_site_margin	0.05075	m
site_drive_margin	162.85197	N
site_turn_time	9.5560071	s
mobility_energy	6231.6801	J
drive_peak_power	544.94266	W
turn_peak_power	197.29647	W
ground_pressure_margin	7940.1632	Pa
straight_motor_margin	510.40004	N
turn_motor_margin	5315.5009	N

Output port	Value	Native units
straight_traction_margin	177.07681	N
turn_traction_margin	-261.1191	N
straight_power_margin	2286.6703	W
turn_power_margin	2536.9756	W
mobility_stability_margin	0.24637225	m
trench_traverse_margin	0.309	m
tip_stability_margin	0.00030000001	m
tip_worst_mask	3	—
obstacle_pose_margin	0.21573457	m
obstacle_tip_margin	0.17631503	m
obstacle_pose_valid	1	—
obstacle_evaluated_height	0.3	m
obstacle_worst_wheel	1	—
obstacle_chassis_pitch	0.30100959	rad
obstacle_chassis_roll	-0.2528901	rad
obstacle_loaded_cg_x	-0.062673061	m
obstacle_loaded_cg_y	0.05051172	m
obstacle_loaded_cg_z	0.52112262	m
obstacle_drum_wheel_gap_bound	0.058025507	m
obstacle_drum_wheel_margin	0.043025507	m
obstacle_clearance_pose_valid	1	—
obstacle_clearance_evaluated_height	0.3	m
slope_pitch_tip_margin	0.0015289604	m
slope_roll_tip_margin	0.00029999998	m
slope_tip_margin	0.00029999998	m
slope_tip_angle	0.52407615	rad
rock_sweep_tip_margin	0.17631503	m
rock_sweep_worst_height	0.3	m
rock_sweep_raised_wheel	1	—
rock_sweep_pitch	0.30100959	rad
rock_sweep_roll	-0.2528901	rad
rock_sweep_cg_x	-0.062673061	m
rock_sweep_cg_y	0.05051172	m
rock_sweep_cg_z	0.52112262	m
rock_sweep_wheel_load_margin	106.3161	N
rock_sweep_load_height	0.3	m
rock_sweep_load_raised_wheel	0	—
rock_sweep_unloaded_wheel	0	—
rock_sweep_pose_valid	1	—
rock_sweep_evaluated_height	0.3	m
rock_sweep_samples	41	—
rock_sweep_force_balance_error	2.5401903e-13	N
empty_slope_slip	0.16683093	—
empty_slope_solution_valid	1	—
empty_slope_traversal_margin	0.83316907	—
empty_slope_force_shortfall	0	N

Output port	Value	Native units
empty_slope_sinkage	0.0054966803	m
empty_slope_transfer_flag_count	0	—
loaded_slope_slip	0.16063554	—
loaded_slope_solution_valid	1	—
loaded_slope_traversal_margin	0.83936446	—
loaded_slope_force_shortfall	0	N
loaded_slope_sinkage	0.012770515	m
loaded_slope_transfer_flag_count	2	—
bottom_pivot_ground_clearance	0.16939994	m
rocker_pivot_ground_clearance	0.18191687	m
ground_clearance	0.16939994	m
ground_clearance_margin	0.016999942	m
bottom_pivot_axle_radius	0.015922205	m
rocker_pivot_axle_radius	0.016865329	m
operating_reach_margin	0.02220294	m
front_dig_actuator_length	0.49488974	m
feed_drive_distance	4.421386	m
axle_ground_margin	0.00015000003	m
front_dig_y	0.1545	m
cut_depth	0.025	m
inlet_width	0.344625	m
rear_dig_x	-0.65100614	m
excavation_domain_margin	0.1545	m
rear_dig_actuator_length	0.49918386	m
front_dig_x	0.65538559	m
rear_dig_y	0.1295	m
front_dig_depth	0.025	m
rear_dig_depth	0.05	m
feed_margin	0.020741024	m
feed_distance	3.1149942	m
feed_per_scoop	0.0042589762	m
feed_speed_limit	0.2	m/s
feed_time	130.02223	s
pair_cutting_time	91.594745	s
soil_supply_rate	0.0011741997	m**3/s
scoop_throughput	0.0068925	m**3/s
feed_drive_energy	1728.1669	J
feed_drive_peak_power	18.549378	W
feed_motor_margin	6938.7375	N
feed_traction_margin	0.3	N
feed_power_margin	2665.6735	W
outbound_travel_time	25.696146	s
return_travel_time	25.696146	s
outbound_route_acceleration	0.4	m/s**2
return_route_acceleration	0.4	m/s**2
outbound_straight_speed_limit	1	m/s

Output port	Value	Native units
return_straight_speed_limit	1	m/s
outbound_bwr_slip	0.22926566	—
return_bwr_slip	0.25522323	—
outbound_bwr_sinkage	0.012879657	m
return_bwr_sinkage	0.0055731064	m
outbound_bwr_transfer_flag_count	2	—
return_bwr_transfer_flag_count	0	—
pit_depth_margin	0.27035	m
transport_clearance_margin	0.30932843	m
wheel_first_margin	0.030900018	m
startup_position_margin	0.0001	m
dig_position_margin	0.098687549	m
pivot_spacing_margin	0.025053015	m
pivot_mount_margin	0.359	m
pivot_longitudinal_margin	0.21761232	m
bottom_mount_margin	0.067122146	m
stroke_margin	0.058787549	m
triangle_retracted_margin	0.027189719	m
triangle_extended_margin	0.054128761	m
toggle_margin	0.42855392	m
wheel_clearance_margin	0.00029999999	m
deployed_height_margin	1.660291	m
actuator_pivot_longitudinal_margin	0.19892662	m
actuator_toggle_margin	0.27889693	m
actuator_catalog_retracted_length	0.52369563	m
actuator_catalog_extended_length	0.68615246	m
actuator_unit_mass	2.4421486	kg
actuator_stroke_domain_margin	0.11165683	m
actuator_force_margin	3.9784037	N
actuator_curve_margin	3.9784037	N
actuator_curve_valid	1	—
actuator_loaded_speed	0.0099080193	m/s
actuator_raising_current	13.990062	A
actuator_lowering_current	14	A
actuator_raising_time	9.9704639	s
plate_mass_total	0.5115886	kg
cross_tube_mass_total	0.50066799	kg
suspension_axle_mass	0.081638358	kg
suspension_bearing_span	0.28891078	m
cross_tube_outer_diameter	0.031844409	m
plate_geometry_margin	9.9999999e-05	m
frame_packing_margin	0.051280388	m
plate_geometry_valid	1	—
structure_cg_z	0.26012356	m
structure_sprung_cg_z	0.2692177	m
structure_yaw_inertia	0.050622695	kg*m**2

Output port	Value	Native units
plate_stress	29806569	Pa
plate_stress_margin	90193431	Pa
plate_buckling_margin	5565.0381	N
cross_tube_bending_stress	1.199e+08	Pa
cross_tube_bending_margin	99999.933	Pa
frame_mount_relative_deflection	0.00050003298	m
rocker_deflection	0.00043994766	m
axle_deflection	9.2638605e-05	m
arm_deflection	0.0010217327	m
rocker_buckling_margin	125333.52	N
arm_buckling_margin	21708.052	N
frame_gravity_actuator_force	5122.3967	N
frame_nominal_wheel_reaction	362.82881	N
frame_force_balance_error	2.6147973e-12	N
frame_load_solution_valid	1	—